School of Computing and Communications

Lancaster University

SCC.160 – Fundamentals of Communication Systems

Tutorial Questions and Revision (Week 23 – Satellite Navigation)

KNOWLEDGE

1. What are the 3 **segments** of GPS?

   The space segment, the control segment and the user segment.

2. What is the **average life expectancy** of a GPS satellite? **How many** satellites comprise the GPS constellation?

   About 10 years. Initially, 24 satellites comprised the GPS constellation. Now there are more than 30.

3. What is the **function** of the **monitoring stations** and the **master control station** of the GPS network?

   The four unmanned receiving stations constantly receive data from the satellites and then send that information to the master control station. The master station 'corrects' the satellite data and sends the information to the GPS satellites.

4. **How many satellites** (in <u>theory</u>) should be in view in order for a GPS receiver to provide satisfactory location information for **civilian** use? Would your answer be different if you considered GPS receivers for **aviation** use?

   For civilian use, readings from three satellites can provide the longitude and latitude of the GPS receiver. For aviation, the altitude is also required; hence four satellites should be in view.

5. Explain how knowledge of local atmospheric conditions can improve GPS accuracy (**assisted GPS**).

   Atmospheric components introduce delay and reduce signal strength. The GPS system uses a built-in model to calculate an average amount of delay to partially correct GPS coordinate estimates but accuracy could still be degraded by up to 10m. Accuracy can be improved by deploying a network of reference stations that monitor local atmospheric conditions, collect data and transmit them to master stations. Master stations generate correction signals and transmit them to GPS receivers via geostationary satellites. Assisted GPS, which was initially developed for civil aviation, yields an accuracy of less than 3m on average.

6. Name **5 sources** of potential **errors**, which affect the **accuracy** of the GPS measurement.

   (i) ionosphere and troposphere delays; (ii) signal multipath; (iii) orbital errors; (iv) receiver clock errors; (v) satellite geometry and shading; (vi) intentional degradation by re-activating selective availability

UNDERSTANDING

1. Movies/TV often show people being followed using a "GPS tracker" or some similar device.
   a. Can this be done? [**Yes** / **No**] (*be prepared to support your answer*)
   b. Could you be traced when you use your in-car sat-nav device? [**Yes** / **No**]
   c. Could you be traced when you use a mobile phone?
   [**Yes** / **Only if a sat-nav application has been installed** / **No**]

   It can only be done if the GPS tracker is both a receiver and a transmitter. A standard sat-nav device can only receive information from GPS satellites, so communication is one-way and tracking cannot be done. When you use a mobile phone, trilateration can pinpoint your location with or without a sat-nav application.